Worksheet 2

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Semester: 4th
Date of Performance: 02/02/2026
Subject Name: Internet of Things Lab
Subject Code: 24CAP-751

1. Aim/Overview of the practical:

Interface an Arduino Uno with Ultrasonic Sensor to calculate the distance and print on a Serial monitor and on LCD. Follow the mentioned cases after the calculation of distance.

a) If the distance is > 250 cm green light is in ON condition.
b) If the distance is < 250 cm and the distance > 90 yellow light is in ON condition and the green light remains Off
c) If the distance is < 90 cm turn on the buzzer and lights will remain OFF.

2. Apparatus

Hardware Requirements

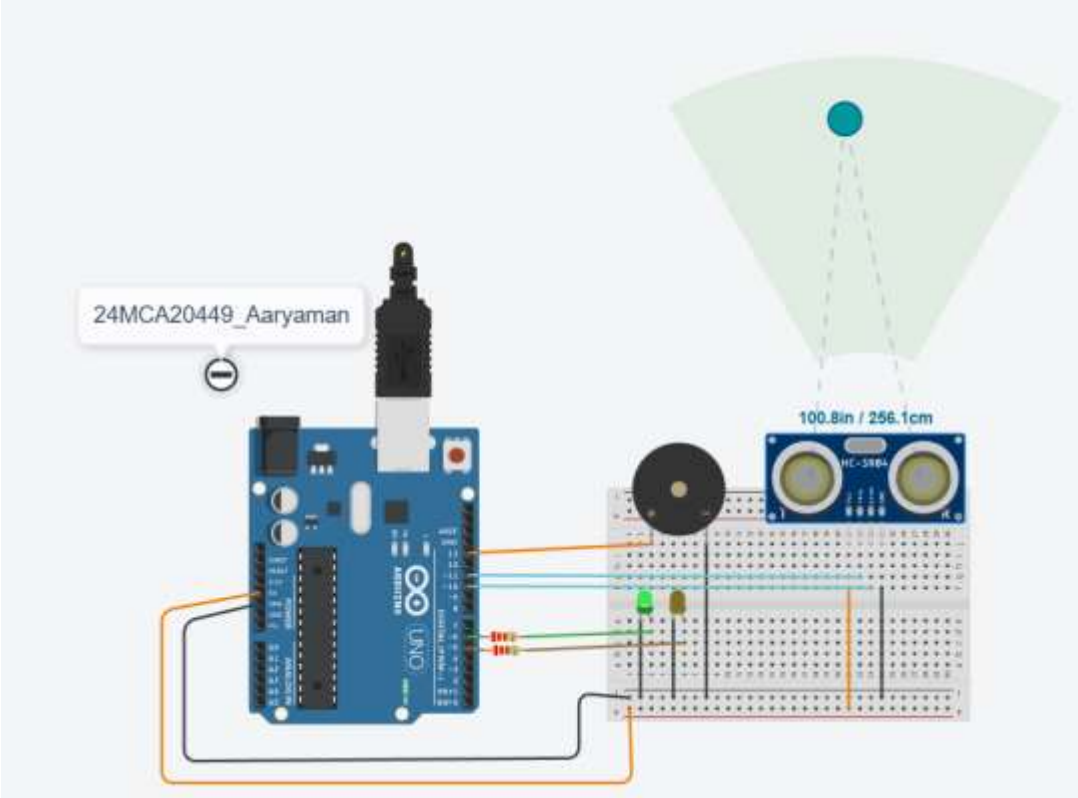
- Arduino Uno
- Breadboard
- Green LED
- Yellow LED
- Ultrasonic Sensor (HC-SR04)
- Buzzer
- 220Ω Resistors
- Connecting Wires

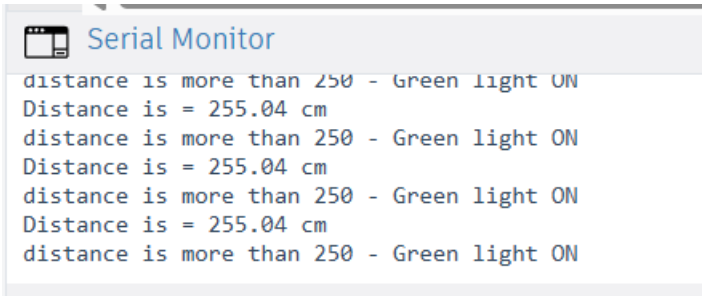
Software requirements

- Tinkercad Circuit
- Arduino programming environment

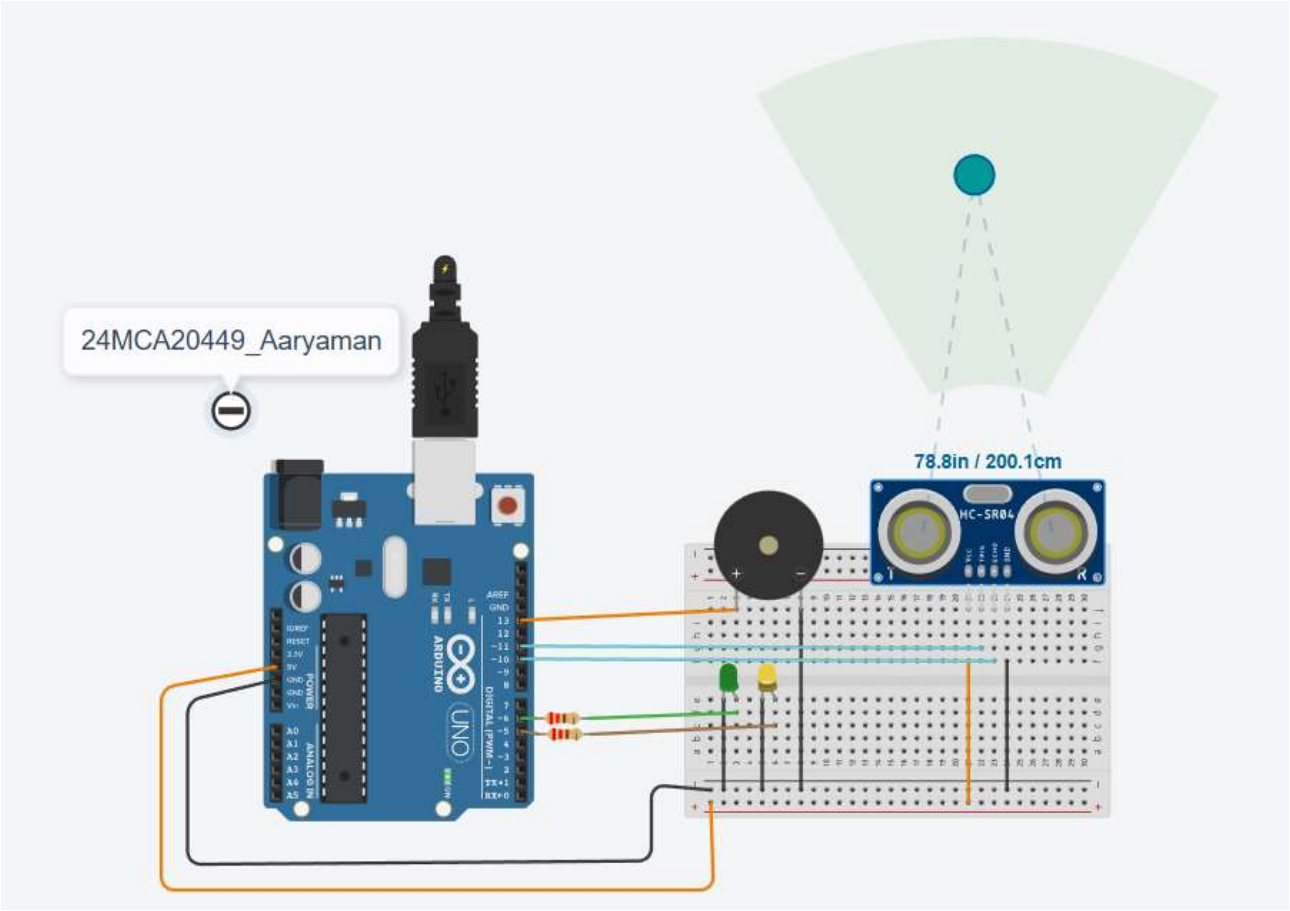
3. Circuit Diagram(TinkerCad):

a) If the distance is > 250 cm green light is in ON condition.





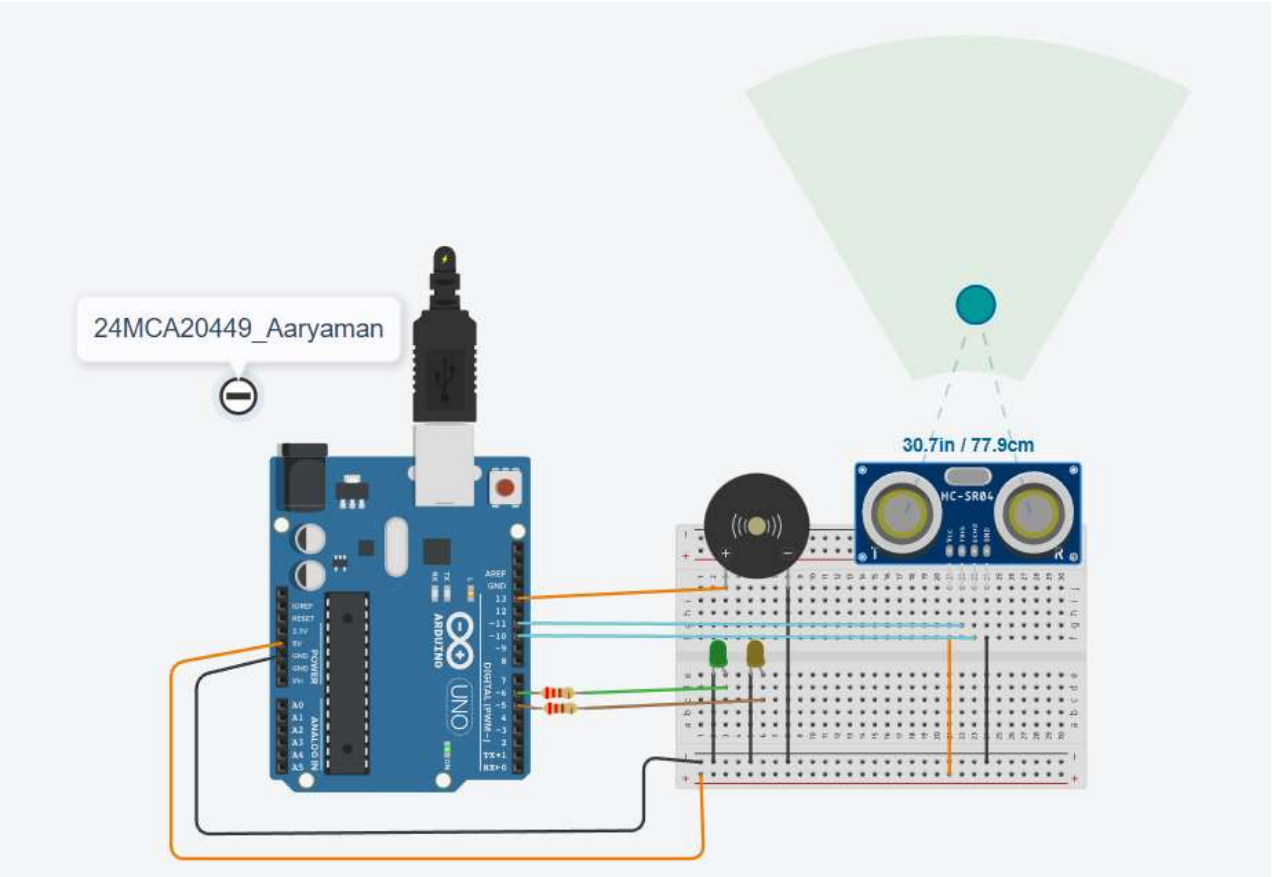
b) If the distance is < 250 cm and the distance > 90 yellow light is in ON condition and the green light remains Off

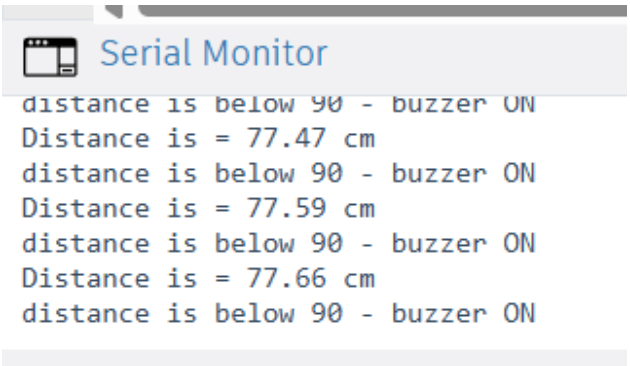


Serial Monitor

```
distance is between 250 to 90 - Yello light ON
Distance is = 199.25 cm
distance is between 250 to 90 - Yello light ON
Distance is = 199.40 cm
distance is between 250 to 90 - Yello light ON
Distance is = 199.25 cm
distance is between 250 to 90 - Yello light ON
```

c) If the distance is < 90 cm turn on the buzzer and lights will remain OFF.





1. Coding:

```
#define trig 11
#define echo 10
#define buzzer 13
#define GreenLed 6
#define YellowLed 5


void setup()
{
  pinMode(trig,OUTPUT);
  pinMode(echo,INPUT);
  pinMode(buzzer,OUTPUT);
  pinMode(GreenLed, OUTPUT);
  pinMode(YellowLed, OUTPUT);

  Serial.begin(9600); // initialize serial monitor
}

void loop()
{

  digitalWrite(trig,0);
  delayMicroseconds(2);

  digitalWrite(trig,1);
  delayMicroseconds(10);

  digitalWrite(trig,0);


 float time = pulseIn(echo,HIGH);
 //distance = 0.0343 cm/micro-sec * time (micro-sec);
 float distance  = 0.0343 * time/2;

 Serial.print("Distance is = ");
 Serial.print(distance);
 Serial.println(" cm");
```

```
  //conditions
  if (distance > 250)
  {
    digitalWrite(GreenLed,1);
    digitalWrite(YellowLed,0);
    digitalWrite(buzzer,0);
    Serial.println("distance more than 250 - Green light ON");
  }
  else if (distance > 90 && distance < 250)
  {
    digitalWrite(GreenLed,0);
    digitalWrite(YellowLed,1);
    digitalWrite(buzzer,0);
    Serial.println("distance between 250 to 90 - Yello light ON");
  }
  else if (distance < 90)
  {
    digitalWrite(GreenLed,0);
    digitalWrite(YellowLed,0);
    digitalWrite(buzzer,1);

    Serial.println("distance below 90 - buzzer ON");

  }

  delay(1000);
}
```

5. Learning outcomes (What I have learnt):

1. Understood the working of HC-SR04 ultrasonic sensor for distance measurement.

2. Learned to interface LEDs and buzzer with Arduino

3. Gained practical knowledge of pinMode(), digitalWrite(), and pulseIn() functions.

4. Developed logic using conditional statements to control outputs based on distance.

5. Improved understanding of proper circuit wiring, common ground, and resistor usage.

Sr. No.	Parameter	Marks Obtained	Maximum Marks
1.	Worksheet Completion		10 marks
2.	Post Lab Quiz Result		5 marks
3.	Student Engagement (Simulation/ Demonstrate/Performance and Pre-Lab Questions))		5 marks
	Total		20 marks